Adult Income Prediction

Overview

This project uses supervised binary classification to predict whether an adult's annual income exceeded $50,000. I trained a logistic regression model using the UCI Adult dataset (Becker & Kohavi, 1996) and compared it with a majority-class baseline. The dataset is available at https://archive.ics.uci.edu/dataset/2/adult.

Dataset Description

Each row represents one adult in an extract from the 1994 Census database (Becker & Kohavi, 1996). The combined CSV has 48,842 rows and 15 columns, including the income target, and 48,790 rows remain after duplicate removal. The predictors include age, education level, occupation, work class, and hours worked per week. About 76% of records are labeled $50,000 or less and 24% are labeled above $50,000.

Modeling Approach

I trimmed whitespace, removed the trailing period from income labels, and mapped the labels to two classes. I treated question marks as missing values, used Unknown for missing categories, and removed 52 duplicate rows after normalizing the labels. I dropped the Census sampling weight (fnlwgt) and the education label because education_num already encodes the level. I also excluded sex and race from the model inputs and used them only for the subgroup audit.

I used a stratified 80/20 split to preserve the class proportions. The scikit-learn pipeline imputes missing values, one-hot encodes categories, and scales numeric inputs so feature units do not drive the effect of regularization. The pipeline fits those steps on training data within each fold, preventing validation and test data from influencing preprocessing (Scikit-learn developers, n.d.).

Logistic regression is an interpretable baseline that estimates the probability of a binary outcome. For Adult, it handles the one-hot encoded categories directly and its coefficients show how individual features contribute to the prediction. It models the log-odds as a linear combination of the predictors (Hastie et al., 2009). The evaluation assumes that rows are sufficiently independent and that the held-out sample represents the same population as the training data. The 0.50 threshold applies a symmetric error-cost rule under calibrated probabilities. The analysis did not establish calibration or measure the real-world costs of the two error types.

A majority-class prediction already achieves 0.761 accuracy because the classes are imbalanced. I therefore judged the model mainly on recall, F1, and ROC-AUC. Precision measures how often a positive prediction is correct, recall measures how many actual positives are found, and F1 is their harmonic mean (Sokolova & Lapalme, 2009). ROC-AUC summarizes ranking across thresholds. The precision–recall curve shows the trade-off for the smaller positive class (Saito & Rehmsmeier, 2015).

Results

The notebook's Task 4, Select and Train a Machine Learning Model, reports five-fold training cross-validation F1 of 0.654 ± 0.011 (mean ± standard deviation). The held-out F1 of 0.671 was close to the training-fold scores, which ranged from 0.640 to 0.672.

The metrics table in Task 5, Evaluate Model Performance, reports test accuracy of 0.855, precision of 0.737, recall of 0.615, F1 of 0.671, and ROC-AUC of 0.909. The majority-class baseline had 0.761 accuracy, zero recall, and a ROC-AUC of 0.500. The confusion matrix shows 899 missed high earners and 512 false alarms. The model missed more high earners than it falsely flagged, so it predicted the lower income category more often than that category occurred. The ROC and precision–recall curves in Task 5 show how performance changes across thresholds.

Interpretation for a Non-Technical Audience

Of every 100 people the model flags as earning over $50,000, about 74 actually do. Of every 100 people who earn over $50,000, the model finds about 62. Guessing that everyone earns $50,000 or less gets about 76 out of 100 people right simply because that group is larger, while missing every high earner. These results describe patterns in historical records. The model should not be used to make decisions about individuals.

Limitations and Potential Bias

The data are from 1994, and the $50,000 cutoff is in nominal dollars. Changes in prices and the labor market limit what the test results tell us about present-day incomes. The model also uses additive effects, so it cannot learn interactions such as different effects of hours worked across occupations unless interaction terms are supplied.

In the Task 5 Fairness and Bias Check, the true positive rate was 0.496 for women and 0.639 for men, a gap of 14.2 percentage points calculated from unrounded rates. I excluded sex and race from the inputs and audited the held-out predictions by group. The relationship field still contains Husband and Wife categories (Becker & Kohavi, 1996), which may act as proxies for sex. Removing a protected attribute can leave correlated information in the remaining features (Mehrabi et al., 2021). The audit did not establish that relationship caused the gap.

The race gap was 15.6 percentage points among groups with at least 30 actual positives and 30 actual negatives. The Other category had only 10 actual positives, so a single case changes its true positive rate by 10 percentage points. As a next experiment, I would use separate validation data to evaluate group-specific thresholds aimed at equalizing recall (Hardt et al., 2016). I would check how the thresholds affect false positives and overall performance on fresh test data.

References

Becker, B., & Kohavi, R. (1996). *Adult* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5XW20

Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. In D. D. Lee, M. Sugiyama, U. V. Luxburg, I. Guyon, & R. Garnett (Eds.), *Advances in neural information processing systems* (Vol. 29, pp. 3315–3323). Curran Associates. https://arxiv.org/abs/1610.02413

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer. https://doi.org/10.1007/978-0-387-84858-7

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, *54*(6), Article 115. https://doi.org/10.1145/3457607

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, *10*(3), Article e0118432. https://doi.org/10.1371/journal.pone.0118432

Scikit-learn developers. (n.d.). *Common pitfalls and recommended practices*. Retrieved November 11, 2026, from https://scikit-learn.org/stable/common_pitfalls.html

Sokolova, M., & Lapalme, G. (2009). A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, *45*(4), 427–437. https://doi.org/10.1016/j.ipm.2009.03.002